8 Projects (optional, for fun)

FULL

n8n 3. A weekly garden report by email

Quick version

No time to read it all? This is the short version. The full guide is below.

- Every Sunday at 8 am, n8n emails you a short report: how many plants your website has, and the current soil moisture.
- Medium, 1 hour, free. Do n8n 1 first, with the garden and Home Assistant running.
- Make a Home Assistant long-lived token. Store it in n8n as a **Header Auth** credential (`Bearer`, then the token). Treat it like a password.
- The workflow: **Schedule Trigger** (Weeks, Sunday, 8:00), two **HTTP Request** boxes (the garden, then Home Assistant), then **Send Email**.
- Click **Execute workflow** once to test, then **Save** and switch it to **Active**.
- No email on Sunday? The workflow must be **Active**. Look in **Executions** for errors.

Level: medium. **Time:** 1 hour. **Cost:** free.

What you will build

Every Sunday at 8 am n8n asks your website how many plants it has, asks Home Assistant for the current soil moisture, and emails you a short report at **you@gmail.com**.

```
Schedule --> ask the garden --> ask Home Assistant --> build a message --> email
```

What you need first

- n8n 1 done.
- The garden website running (guide 03) and Home Assistant running (Project 3).
- A Home Assistant **long-lived token**: HA, click your name (bottom left), **Security**, **Long-lived access tokens**, **Create token**. Copy it once. The token lets n8n read Home Assistant. Treat it like a password.

Steps

Part A. Give n8n the Home Assistant token safely

1. In n8n: **Credentials, Create credential**, search **Header Auth**.
2. Name: `Home Assistant token`. Header Name: `Authorization`. Value: `Bearer` followed by your token (the word Bearer, a space, then the token).
3. Save. The token is now stored inside n8n, not in the workflow.

Part B. The workflow

4. **Create workflow**, name it `Weekly garden report`.
5. **Add first step: Schedule Trigger.** Trigger interval: **Weeks**. Day: Sunday. Hour: 8. Minute: 0. (Uses the time zone from `docker/extras/n8n.yml`.)
6. Add **HTTP Request**. Name it `Ask the garden`. Set: - Method: GET - URL: `http://127.0.0.1:8030/healthz` This asks the website for its plant count. (n8n runs on the laptop's own network, so `127.0.0.1` means "this laptop" and reaches your garden.)
7. Add another **HTTP Request**. Name it `Ask Home Assistant`. Set: - Method: GET - URL: `http://127.0.0.1:8123/api/states/sensor.garden_bed_1_moisture_percent` - Authentication: **Generic Credential Type**, Generic Auth Type: **Header Auth**, credential: `Home Assistant token`
8. Add **Send Email**. Credential `My Gmail (send)`; From and To `you@gmail.com`; Subject: `Your weekly garden report`. Text (choose **Expression**):

```
Good morning!

Plants in the catalogue: {{ $('Ask the garden').item.json.plant_count }}
Bed 1 soil moisture: {{ $('Ask Home Assistant').item.json.state }} percent

Have a good week in the garden.
```

9. Click **Execute workflow** once to test. *You should see:* green ticks and an email.
10. **Save** and switch the workflow to **Active**.

How you know it worked

The test email arrives. The next Sunday at 8 am another one arrives by itself.

If something goes differently

What you see	What to do
`ECONNREFUSED` for port 8030	The garden is not running. `docker compose up -d` in the project folder, then `curl http://127.0.0.1:8030/healthz`.
`ECONNREFUSED` for port 8123	Home Assistant is not running: `bash lab/07-start-extras.sh ha`.
n8n will not open at `localhost:5678`	`docker logs garden-n8n` shows why. Check that no other program uses port 5678: `sudo ss -ltnp` (look for :5678).
401 from Home Assistant	The token is wrong or has no `Bearer` in front. Make a new token.
`state` is `unavailable`	The sensor is offline. Check Project 6.
No email on Sunday	The workflow must be **Active**. Check **Executions** (left menu) for errors.

Safety and best practice

- The token is in n8n's credentials, not in the workflow text. If you export or share a workflow, credentials are not included. Check anyway before sharing.
- Back up `n8n-data/` with the rest of your lab.

Make it yours

- Add the weather forecast from Project 5 to the email.
- Add the last watering time and how many minutes the valve ran.
- Send the report to a second person too.
- Use an **Email Trigger (IMAP)** box (host `imap.gmail.com`, port 993, SSL on, same app password in a new IMAP credential) to react to emails you send yourself, for example a subject of `water bed 1`. Ask Gemini to plan it first, and add a safe limit so a stranger's email cannot start the water.

Back to the plan.